

**DESIGN AND FABRICATION OF A 3-IN-1
MULTIPURPOSE PIGGY BANK**

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report “**DESIGN AND FABRICATION OF A 3-IN-1 MULTIPURPOSE PIGGY BANK**” is the bonafide work of “**GIRI R (Register No: 212223060068)**” who carried out the project work under my supervision.

ABSTRACT

This project presents the planning, design, and step-by-step fabrication of a compact multipurpose wooden box that combines three everyday utilities into a single unit: a coin-and-currency piggy bank, a pin/needle stand, and a key-chain hanger. The enclosure is built primarily from plywood, with a hasp lock, a hinge, and screw-eye hooks providing the mechanical function of each feature. A circular hole on the front face allows pins and needles to be dropped in and stored upright, two horizontal slots below it allow coins and folded currency notes to be deposited, and three hooks on the top face allow keys or key-chains to be hung. The lid is secured with a hasp-and-staple lock, optionally paired with a small padlock, protecting the stored contents.

The report documents the design requirements, the materials and tools required, and an eight-step fabrication procedure — marking and cutting the panels, assembling the box frame, drilling the pin-stand hole, cutting the coin/note slots, fixing the lid hinge, fixing the hasp lock, fixing the key-chain hooks, and sanding and finishing — each illustrated with a schematic diagram. Photographs of the completed model are presented, showing the pin-stand hole, coin/note slots, hasp lock, and key-chain hooks in place, with the remaining faces left as plain, unfinished plywood.

The completed model was found to satisfy all three intended functions — secure money storage, a dedicated pin stand, and a key-chain hanger — within a single, low-cost enclosure, demonstrating that multiple everyday utilities can be combined into one compact desk accessory using simple hand-tool fabrication.

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LIST OF SYMBOLS AND ABBREVIATIONS

ECE – Electronics and Communication Engineering

PVA – Polyvinyl Acetate (wood adhesive)

mm – Millimetre

cm – Centimetre

Ø – Diameter

1. INTRODUCTION

1.1 General

Piggy banks, pin cushions, and key holders are common small household items that are usually bought and kept separately. This project combines all three functions into one compact plywood enclosure in order to reduce clutter and cost, while practising basic carpentry skills such as marking, cutting, drilling, and fitting hardware.

1.2 Objectives of the Project

- To design a single enclosure that performs three functions — money storage, a pin/needle stand, and a key-chain hanger.
- To fabricate the enclosure using low-cost, easily available materials (plywood and small metal hardware).
- To secure the stored contents using a lockable lid.
- To document the fabrication process step by step, with a diagram for each step, for reproducibility.

2. PLANNING AND DESIGN

2.1 Design Requirements

- A lockable compartment to store coins and folded currency notes.
- A separate opening for dropping in pins/needles so they can be stored upright, without mixing with the money compartment.
- An external fitting to hang keys or key-chains, so the unit can double as a key holder.
- Overall size small enough to sit on a table or desk.
- Use of materials that are cheap, light, and easy to work with by hand.

2.2 Concept and Working Principle

The design places all three functions on two faces of the same plywood box, as shown in Figure 2.1.

- Pin-Stand Hole — a circular hole on the front face lets pins/needles be dropped in vertically; they collect inside the box and can be tipped or fished out when needed.
- Coin & Note Slots — two horizontal slots below the hole allow coins and folded notes to be pushed in; they collect inside the same locked enclosure.
- Key-Chain Hooks — three screw-eye hooks fixed on the top face let keys or key-chains be hung when not in use.
- Lock — a hasp-and-staple latch on the lid, with an optional small padlock, keeps the box closed and the stored money secure.

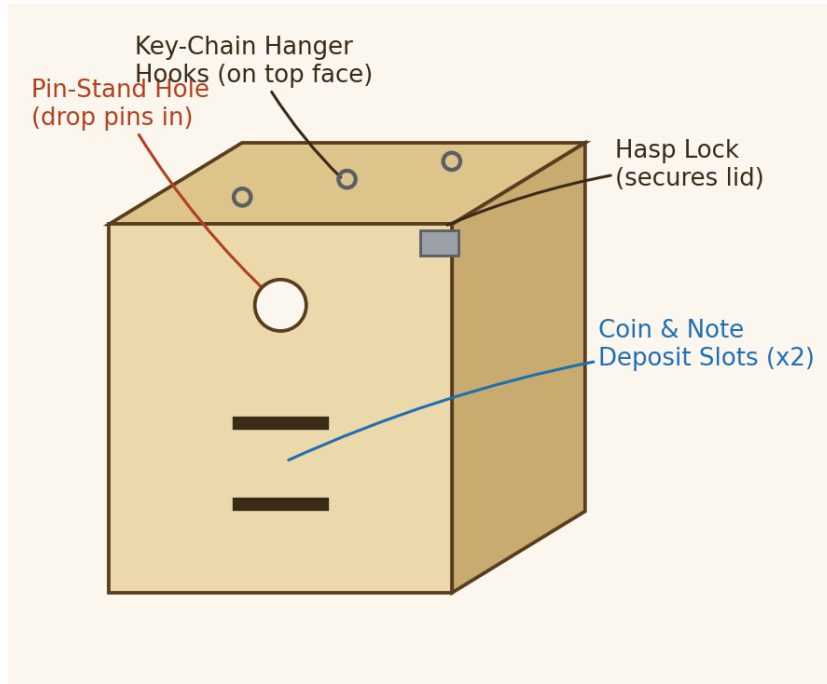


Figure 2.1 Conceptual Design of the Three Functions

2.3 Materials and Tools Required

Based on the hardware visible in the finished unit and standard practice for this kind of build, the following materials and tools are required.

Table 2.1 Materials Required

Material	Approx. Qty / Spec	Purpose
Plywood sheet (≈ 6 mm)	6 panels (front, back, lid, bottom, 2 sides)	Main body of the box
Wood adhesive (e.g., PVA)	As required	Bonding panel joints
Panel pins / small nails	$\approx 20-25$	Reinforcing glued joints
Hasp & staple lock	1 set	Locking the lid
Small hinge	1	Allows the lid to open and close
Screw-eye hooks	3	Key-chain hanger

Material	Approx. Qty / Spec	Purpose
Sandpaper (medium & fine grit)	As required	Smoothing edges and surface
Small padlock (optional)	1	Extra security for the hasp lock

Table 2.2 Tools Required

Tool	Purpose
Measuring tape / ruler & pencil	Marking panel dimensions and cut-lines
Hand saw / coping saw or rotary cutting tool	Cutting panels and slots
Electric drill with hole-saw / spade bit	Drilling the pin-stand hole
Screwdriver	Fixing the hinge, hasp lock, and hooks
Hammer	Driving panel pins
Sanding block	Finishing edges and surfaces

2.4 Approximate Dimensions

Table 2.3 Approximate Dimensions

Parameter	Approximate Value
Overall box size (L x W x H)	≈ 15 x 15 x 18 cm
Pin-stand hole diameter	≈ 25–30 mm
Coin/note slot size	≈ 55–60 mm long x 4–5 mm wide
Plywood thickness	≈ 6 mm

Note: dimensions are indicative and should be adjusted to the plywood off-cuts on hand and personal preference.

3. FABRICATION PROCEDURE

The steps below outline the general fabrication sequence, each illustrated with a schematic diagram of that stage.

3.1 Step 1: Marking and Cutting the Panels

Using the required dimensions, mark six panels — front, back, lid/top, bottom, and two sides — on the plywood sheet with a pencil and ruler, then cut them out using a hand saw or coping saw.

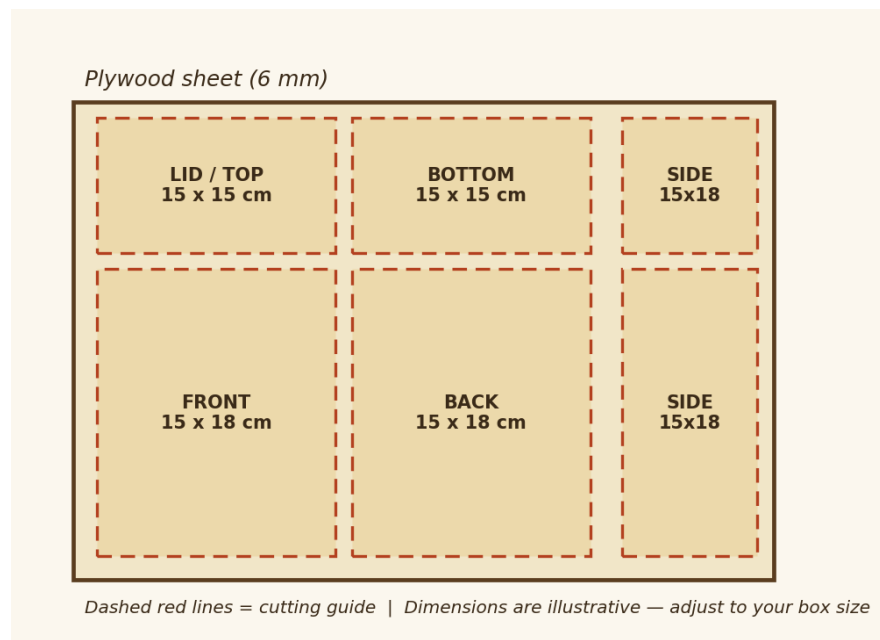


Figure 3.1 Marking and Cutting the Panels

3.2 Step 2: Assembling the Box Frame

Join the back, the two side panels, and the bottom panel using wood adhesive reinforced with panel pins to form an open box shell. The lid panel is kept separate at this stage so it can be hinged in Step 5, and the front panel is fixed only after the hole and slots (Steps 3–4) have been cut into it.

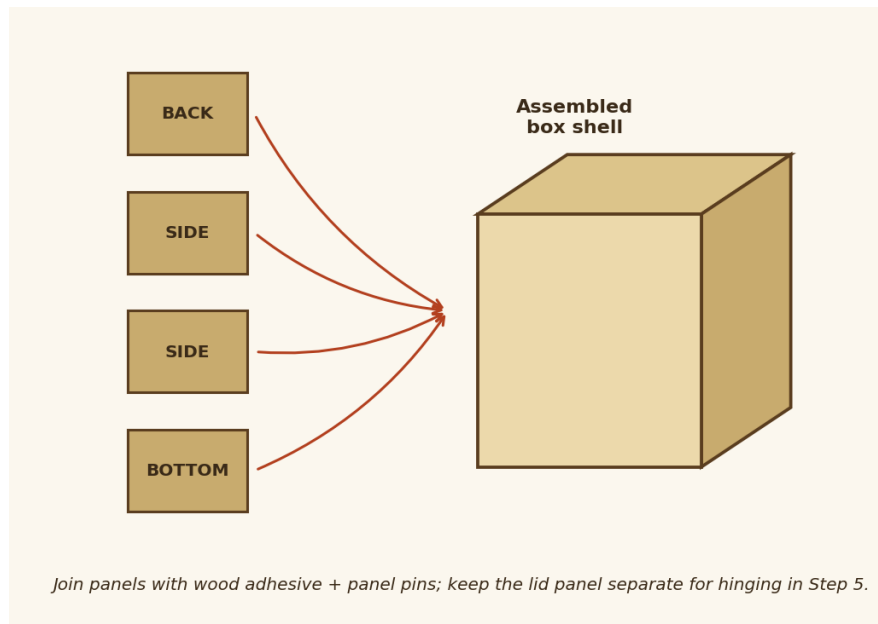


Figure 3.2 Assembling the Box Frame

3.3 Step 3: Marking and Drilling the Pin-Stand Hole

On the front panel, mark the centre of the pin-stand hole near the top and drill a circular hole ($\approx 25\text{--}30\text{ mm}$ diameter) using an electric drill with a hole-saw or spade bit.

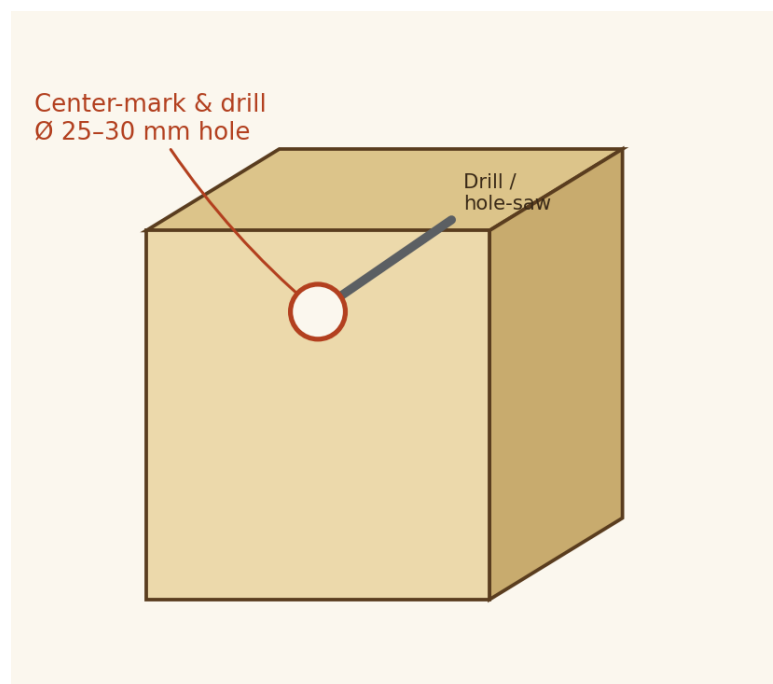


Figure 3.3 Marking and Drilling the Pin-Stand Hole

3.4 Step 4: Marking and Cutting the Coin/Note Slots

Below the pin-stand hole, mark two horizontal slots ($\approx 55\text{--}60$ mm long, 4–5 mm wide) and cut them using a fine saw or rotary tool — these serve as the coin-and-note deposit openings. Once cut, fix the front panel onto the box shell.

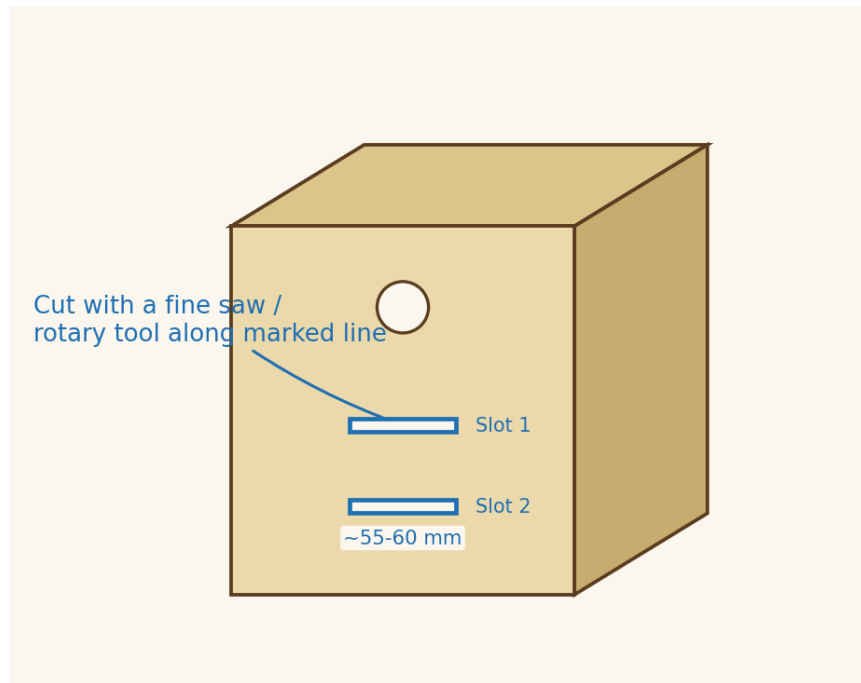


Figure 3.4 Marking and Cutting the Coin/Note Slots

3.5 Step 5: Fixing the Lid Hinge

Attach a small hinge along the back-top edge joining the lid panel to the back panel, allowing the lid to swing open and closed.

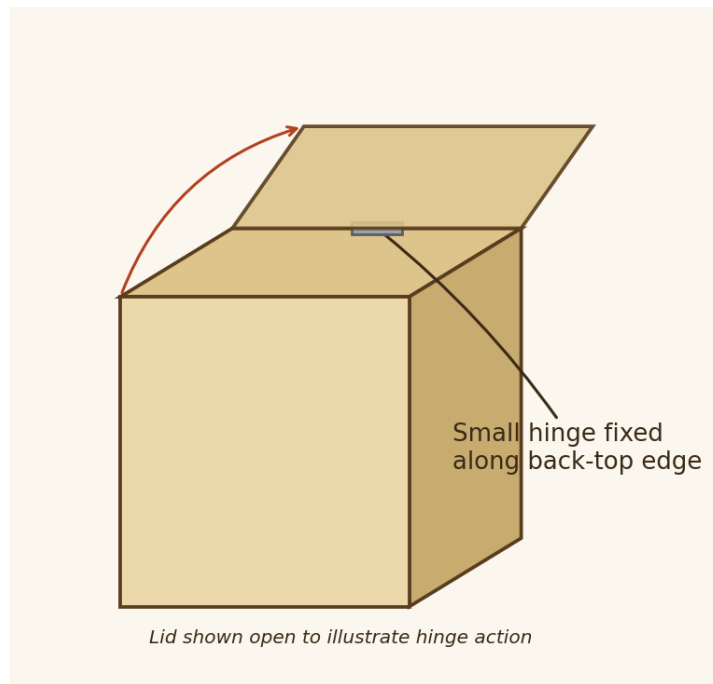


Figure 3.5 Fixing the Lid Hinge

3.6 Step 6: Fixing the Hasp Lock

Fix the hasp-and-staple lock at the joint between the lid and the front panel. A small padlock can be passed through the staple for extra security.

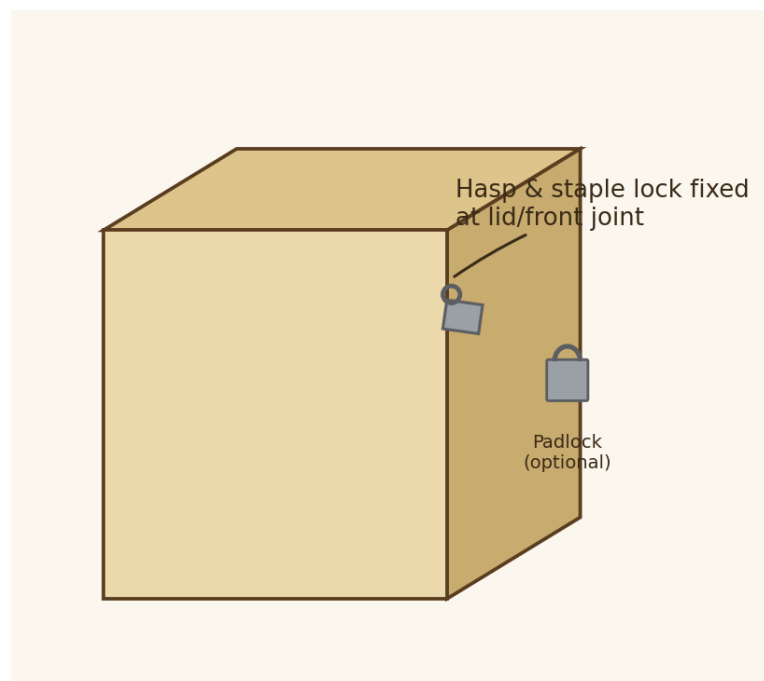


Figure 3.6 Fixing the Hasp Lock

3.7 Step 7: Fixing the Key-Chain Hanger Hooks

Mark three equally spaced points along the top face of the lid and drive in the screw-eye hooks that serve as key-chain hangers.

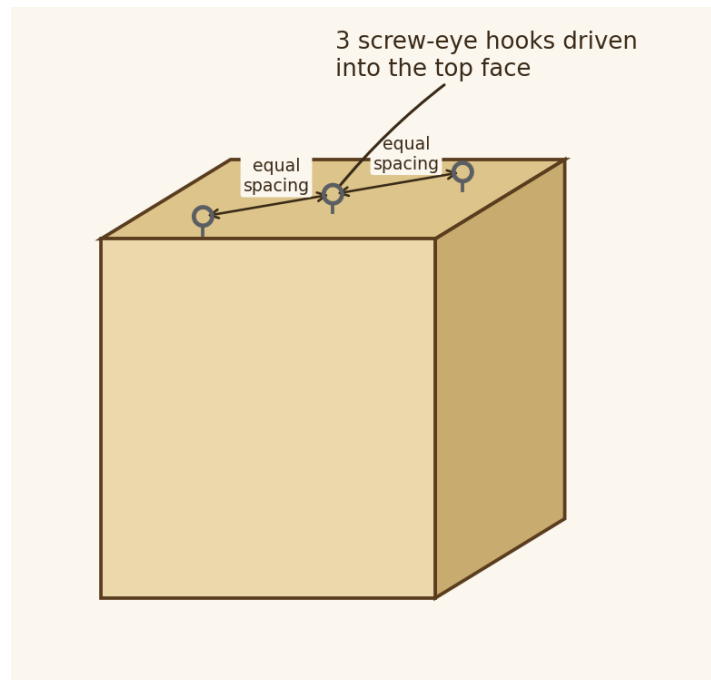


Figure 3.7 Fixing the Key-Chain Hanger Hooks

3.8 Step 8: Sanding, Cleaning and Finishing

Sand all edges and surfaces with medium and then fine sandpaper to remove roughness and splinters, then wipe the unit clean. An optional coat of paint or varnish can be applied for a better finish.

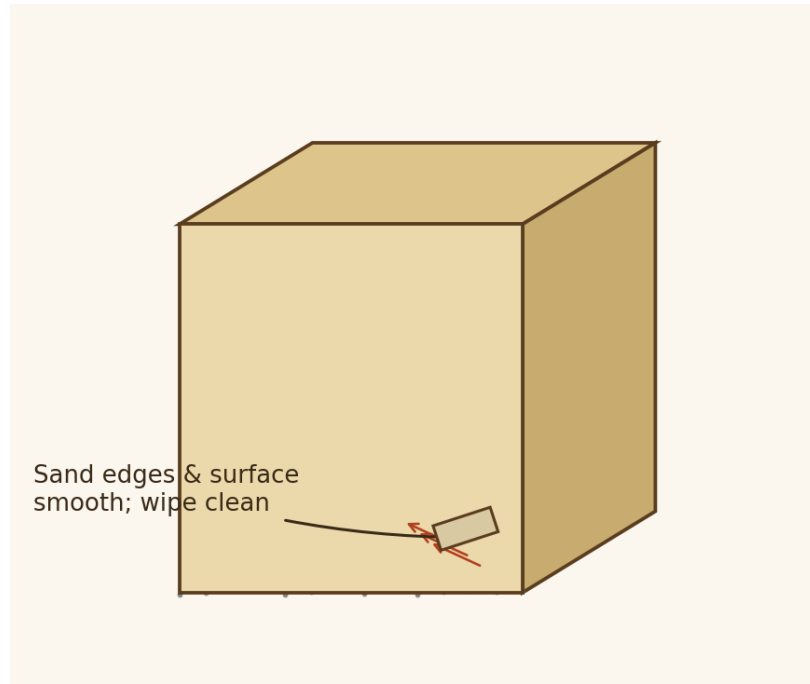


Figure 3.8 Sanding, Cleaning and Finishing

4. RESULTS AND DISCUSSION

4.1 Photographs of the Completed Model

The completed unit was photographed from multiple views. The pin-stand hole, coin/note slots, and hasp lock are visible on the functional front face; the key-chain hooks are fixed on the top face. The remaining faces (back, bottom, and the unshown side) are left as plain, unfinished plywood with no additional features.



Figure 4.1 Top View – Key-Chain Hanger Hooks



Figure 4.2 Side View – Hasp Lock Mechanism



Figure 4.3 Front / Isometric View – Complete Assembly

4.2 Observations

- The circular hole on the front face accepts pins/needles dropped in vertically, providing a dedicated pin-stand separate from the money compartment.
- The two horizontal slots below it accept coins and folded currency notes, which collect inside the locked enclosure.
- The hasp-and-staple lock, visible at the lid joint, secures the lid; paired with a padlock it restricts access to the stored contents.
- The three screw-eye hooks on the top face allow keys or key-chains to be hung, adding the third function to the same unit.
- The box is compact and portable, and built from low-cost plywood and hardware, keeping the overall fabrication cost low.

4.3 Advantages

- Combines three separate household items into one unit, saving desk/table space.
- Simple hand-tool fabrication — no specialised machinery required.
- Low material cost using plywood off-cuts and inexpensive hardware.

4.4 Limitations and Scope for Improvement

- A basic hasp lock is not tamper-proof; a concealed or key-based lock would improve security.
- Untreated plywood is not moisture-resistant; a coat of varnish/paint would improve durability.
- Coin retrieval requires opening the lid — a removable base panel could allow emptying without unlocking.

5. CONCLUSION

The planning and fabrication of a compact 3-in-1 plywood unit — combining a coin-and-note piggy bank, a pin stand, and a key-chain hanger — was completed successfully. The design requirements set out in Section 2.1 were met: the unit stores money securely behind a lockable lid, provides a separate pin-stand opening, and offers a top-mounted key-chain hanger, all within a single low-cost enclosure built from plywood and simple hardware. The step-by-step procedure in Chapter 3 documents the build for reproducibility, and the observations in Chapter 4 confirm that all three intended functions are achieved in the finished model.

APPENDIX 1

VIDEO DEMONSTRATION

A short video demonstrating the fabrication process and/or the working demo of the unit is linked below.

Video Link:

<https://drive.google.com/file/d/18gqgi97oIbMaHWVwEbflGMX7rNFBzKgZ/view?usp=drivesdk>
